

MIA MORRISON



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LinkedIn

PROFESSIONAL SUMMARY

Aerospace Engineering student with experience in mechanical design, robotics, systems integration, and technical project development. Recognized statewide leader and two-time Florida Youth of the Year with a proven record of developing programs, leading teams, organizing large-scale initiatives, and communicating complex ideas to diverse audiences. Passionate about applying systems thinking, engineering design, and leadership to solve real-world challenges.

EDUCATION

EMBRY-RIDDLE AERONAUTICAL UNIVERSITY (ERAU) | B.S. Aerospace Engineering | GPA: 3.33

Est. May 2027

UNIVERSITY OF SUNSHINE COAST (UNISC)

Queensland, Australia – Summer 2024

Study Abroad: Economics and Australian Social Studies

ENGINEERING WORK EXPERIENCE

DOCKING ROBOTICS | MECHANICAL ENGINEERING INTERN

May 2025 – Sept 2025

- Designed and 3D-printed a modular drone landing dock in SolidWorks using a six-panel interlocking system optimized for strength, precision, and rapid assembly.
- Conducted material selection, stress analysis, and torque calculations to improve mechanical performance; currently integrating servo-based rotational actuation for automated movement.

TECHNICAL PROJECTS

PATHFINDER AODR-01 AUTONOMOUS ROVER | Arduino, CAD, 3D Printing

May 2026 – June 2026

- Designed, manufactured, and tested an autonomous obstacle-avoidance rover using an Arduino Nano, ultrasonic sensing, DC motors, LCD display, and custom 3D-printed chassis, while integrating mechanical, electrical, and software subsystems.
- Independently learned Arduino programming, embedded systems, electronics integration, & GitHub documentation.

MATLAB MEAL PREP SYSTEM | MATLAB, Excel

February 2024 – April 2024

- Developed a MATLAB-based application that recommends high-protein, low-calorie recipes based on user meal preferences, protein selection, and available cooking time.
- Designed and integrated a custom Excel-based recipe database and filtering logic to generate personalized recipes.

CATIA SCOOTER DESIGN | CATIA

Sept 2023 – Dec 2023

- Reverse-engineered and modeled a complete electric scooter assembly using CATIA.
 - Created detailed engineering drawings, technical documentation, and assembly models.
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SKILLS

- Software:** SolidWorks, Onshape, CATIA, MATLAB, Arduino IDE, Python, Microsoft 365
 - Leadership & Advocacy:** Public Speaking, Project Management, Team Leadership, Youth Mentorship, Community Engagement
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LEADERSHIP & COMMUNITY EXPERIENCE

FAITH IN FLORIDA- INDIVIDUAL DONOR COORDINATOR | CIVIC ENGAGEMENT ORGANIZER

April 2021- Present

- Lead donor stewardship initiatives, fundraising campaigns, and community engagement efforts for statewide civic programs.
- Organized candidate forums, voter registration drives, teach Black history and college readiness to youth throughout Florida.

NATIONAL PUBLIC SPEAKER, YOUTH ADVOCATE (BGCA)

June 2022-Present

- Advocated for mental health awareness and educational resources nationally as the 2022 & 2023 Florida Youth of the Year.
- Collaborated with Ashley Biden and Congress on Capitol Hill; independently developed a college readiness program.
- Sole Youth Keynote Speaker at a NASCAR Business event, presenting before Dr. Mae Jemison, earning a NASCAR internship.